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# TOAI-elegant-lang: A Tri-Script Orthogonal RISC Wire Protocol, Adversarial Corpus Audit, and Empirical Comprehension Harness for Multi-Agent Communication

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**Repository & Reproducibility Artifact:** https://github.com/PhenoX-AI-Alliance/TOAI-elegant-lang | **PyPI:** `pip install toai-elegant-lang==5.4.4`

**ABSTRACT**

Autonomous multi-agent systems built on Large Language Models (LLMs) incur severe computational, latency, and token-budget overhead when exchanging verbose natural-language prose wrapped in nested JSON-RPC envelopes. Compressing agent-to-agent (A2A) messages exposes a fundamental systems trilemma among *wire token efficiency*, *syntactic, numeric, and causal execution fidelity*, and *human auditability*. This paper presents **TOAI-elegant-lang (v5.4.4)**, a zero-dependency, deterministic compiler and wire protocol inspired by the Rosetta Stone that aligns multi-agent state transitions across three synchronized representations: (1) **Demotic Wire Script**, a compact causal wire syntax (`@CTX:... :: §S01 ... §X64`); (2) **Sacred Expanded Script**, a deterministic macro- and primitive-expanded canonical form (`§GAIA_WARMTH_HARMONY`); and (3) the **Rosetta Audit Layer**, which attaches natural-language definitions and domain invariants for human inspection. Under `o200k_base` and `cl100k_base` Byte-Pair Encoding (BPE) tokenizers, each canonical §-prefixed identifier strictly occupies 3 BPE tokens (`'§' + Prefix + Digits`), while the 256-primitive + 256-macro codebook fits inside a $\sim$1,000-token system prompt header. Evaluated across standard A2A JSON serializers (`benchmarks/fair_bench.py`), Demotic Wire averages **73.2 BPE tokens/message**, achieving a **79.1% token reduction vs. Verbose A2A JSON** (349.8 tokens), **55.7% vs. Structured Tool JSON** (165.2 tokens), and **41.0% vs. Minified Key JSON** (124.2 tokens) while maintaining **100.0% downstream AST/task execution accuracy** (outperforming `LLMLingua`-style perplexity token pruning at $\rho = 0.35$, where numeric/task recovery degrades to 61.6%) and breaking even within **4–20 messages uncached** or **1–2 messages under 90% prompt caching**. We introduce a non-circular **Independent Corpus Alignment Auditor** (`src/audit_corpus_alignment.py`) validated against a human gold-standard sample ($n = 60$, $p_o = 93.33\%$, **Cohen's** $\kappa = 0.8611$). To eliminate circular validation, receiver comprehension is evaluated on a **strictly disjoint** $N = 112$ **Holdout Benchmark** (`corpus/holdout_comprehension.jsonl`, verified at **0.0% wire/scenario overlap**): Sacred expansion raises 4-choice blind accuracy from **44.6% (50/112)** on Compact Wire (Cond A) to **84.8% (95/112)** on Sacred Names (Cond B, exact McNemar $p = 1.36 \times 10^{-9}$, OR = 7.43, Cohen's $h = +0.878$) and **93.8% (105/112)** with Rosetta Glossary (Cond C, $p = 6.35 \times 10^{-3}$ vs. B), while the length-matched Shuffle Control negative baseline collapses to **24.1% (27/112)** (chance = 25.0%, $p = 3.59 \times 10^{-17}$ vs. B). Finally, we formalize origin attestation directly on Page 3 with a game-based EUF-CMA reduction bounding $k$-turn internal-counter HMAC-SHA256 forgery by $10^{-4k} + \epsilon_{\text{PRF}} + \epsilon_{\text{coll}}$.

## 1. Introduction and Multi-Agent Communication Trilemma

As Large Language Model (LLM) applications evolve from single-turn assistants into multi-agent workflows—e.g., Anthropic Model Context Protocol (MCP) [1], Google Agent-to-Agent (A2A) JSON-RPC 2.0 orchestrators [2], and AutoGen [5]—inter-agent communication overhead has become the dominant bottleneck in latency, context saturation, and KV-cache footprint. In conventional topologies, agents coordinate state transitions by generating conversational prose inside nested JSON schemas. While human-readable, schema keys and punctuation delimiters waste token budgets and scale attention FLOPs quadratically. Compressing inter-agent messages exposes a strict **Multi-Agent Communication Trilemma**:

1. **Wire Token Efficiency:** Minimizing BPE subword tokens per transition and amortizing shared schemas across sessions.
2. **Syntactic, Numeric, and Causal Execution Fidelity:** Preserving structured parameters (e.g., leading-zero strings `"007"`, embedded commas `"a, b"`, floats `150.50`), transition probabilities $p \in [0, 1]$, costs $c \in [0, 0.9999]$, and fault edges (`!->`) for downstream AST execution.
3. **Human Auditability and Cross-Model Portability:** Allowing humans and heterogeneous LLM families (GPT, Claude, Gemini, Llama) to inspect and exchange packets over standard text APIs without opaque binary blobs or model-specific latent adapters.

Existing approaches sacrifice at least one vertex (Table 1). Perplexity-based prompt compression (LLMLingua [4]) and progressive pruning (e.g., SafeSieve) drop tokens dynamically, but break JSON syntax delimiters and drop numeric thresholds at high compression ratios ($\rho \leq 0.35$, Section 5). Conversely, emergent MARL discrete codes [6] or continuous latent vectors [7] sacrifice human auditability and fail across commercial text-only APIs. State-delta diffs reduce payload size on homogeneous repeated states but introduce desynchronization risks under network retries and still pay JSON overhead on multi-domain handoffs; crucially, TOAI-elegant-lang is stateless on the wire and orthogonal to state-delta encoding (deltas can be nested directly within Demotic `[k:v]` payloads). Flat key-value templates (`KEY=VAL`) reduce raw tokens but destroy multi-branch causal execution graphs and fault-recovery semantics.

To address this trilemma, we present **TOAI-elegant-lang (v5.4.4)**, a zero-dependency Python compiler and wire protocol built on an orthogonal Reduced Instruction Set Computer (RISC) architecture and Tri-Script Rosetta design. Crucially, release `v5.4.4` incorporates multi-AI adversarial audits (ChatGPT and Claude) and external peer review, replacing provisional estimates with a strictly disjoint $N = 112$ holdout benchmark, complete downstream execution accuracy comparisons, and a formal game-based cryptographic reduction.

**Table 1:** Architectural Comparison of Multi-Agent Communication Paradigms (Verified Standards & Repository Baselines)

| Communication Paradigm | Wire Encoding | Causal AST & Guards | Deterministic Round-Trip | Cross-API Portability | Human Auditability & Operational Characteristics |
|---|---|---|---|---|---|
| Verbose A2A / MCP JSON-RPC [1, 2, 5] | Pretty JSON + NL Prose | Yes (Verbose) | 100.0% | 100% (Text API) | High readability; highest token cost (349.8 tok/msg; baseline). |
| Structured Tool JSON (`fair_bench.py`) | Standard JSON Schema | Yes (JSON Objects) | 100.0% | 100% (Text API) | Lossless schema; repeated JSON keys/syntax (165.2 tok/msg). |
| Minified Key JSON (`fair_bench.py`) | Short-Key JSON (`"c","s"`) | Yes (Abbrev. JSON) | 100.0% | 100% (Text API) | Abbreviated keys require schema lookup (124.2 tok/msg). |
| Simplified Flat Template (`KEY=VAL`) | Delimited Flat String | **No (Lossy)** | Lossy (0% AST) | 100% (Text API) | Lowest raw tokens (34.5 tok/msg), but loses causal edges & 75% params. |
| Binary Protobuf [8] (Base64 Text Wire) | Base64-Armored Octets | Schema-Defined | 100.0% (External) | Requires Tool/Decode | Opaque Base64 string in prompt context; BPE subword fragmentation. |
| Perplexity Token Pruning (LLMLingua [4]) | Pruned Natural Text | Unstructured | Lossy (0% JSON) | 100% (Text API) | Designed for prose/RAG; breaks JSON syntax; 61.6% task acc at $\rho = 0.35$. |
| Emergent MARL Codes [6, 7] | Learned IDs / Vectors | Implicit in Policy | Model-Dependent | Homogeneous Only | Opaque latent symbols; not directly interpretable by humans. |
| **TOAI-elegant-lang (v5.4.4 Ours)** | **Demotic RISC Wire (`§ID`)** | **Yes (Native LL(1) AST)** | **100.0% Lossless** | **100% (Text API)** | **Tri-Script (Demotic / Sacred / Rosetta); 73.2 tok/msg; 100% Task Acc.** |

## 2. System Architecture, LL(1) Grammar, and Quote-Aware Stateful Codec

### 2.1 Tri-Script Architecture and 8-Bit Orthogonal RISC Lexicon Hierarchy

TOAI-elegant-lang decouples machine transmission from human inspection via three synchronized layers (Table 2):

1. **Demotic Wire Script (Compact Machine Layer):** Transmitted between agents. Combines context header (`@CTX:DOMAIN`), 1-byte logical core opcodes (`§S01`–`§H32` across 8 categories: System, Data, Governance, Fault, Execution, Audit, Gaia, Logic; exactly 256 primitives in `core_lexicon.json`), 256 active higher-order macros (`§X01`–`§X969` in `macro_lexicon.json`), typed payloads (`[k:v]`), and causal operators (`-[p:..., c:...]->`, `=>`, `!->`).
2. **Sacred Expanded Script (Canonical Symbolic Layer):** Deterministic expansion via `codec.decode(wire)` expanding opcodes to canonical uppercase snake-case identifiers matching strict `^§[A-Z][A-Z0-9_]*$` (e.g., `§GAIA_WARMTH_HARMONY`).
3. **Rosetta Audit Layer (Human Governance Layer):** Maps every symbol to concise natural-language definitions for compliance auditing (`rosetta_layer.json`) and Condition C glossary injection.

**Exact BPE Token Accounting:** Under `o200k_base` and `cl100k_base`, the section sign `'§'` (`U+00A7`) is tokenized separately from following ASCII characters. In all reported benchmarks, every canonical §-prefixed opcode (`§S01`, `§G48`, `§X64`) is strictly counted as **3 BPE tokens** (`'§'` + `Prefix` + `Digits`), confirming that our 73.2 tok/msg average includes full token penalties.

**Table 2:** Tri-Script Representation and Lexicon Hierarchy in TOAI-elegant-lang v5.4.4

| Architectural Layer | Source / Capacity | Canonical Pattern / Example | Validation & Runtime Role in v5.4.4 |
|---|---|---|---|
| 1. Demotic Wire Script (Core Primitives) | `core_lexicon.json` (256 Primitives) | `§S01..§L32`, `§G48`, `§C20` (3 BPE tok/symbol) | Immutable 256-primitive base vocabulary; validated at startup and by `test_17` for bijective mapping. |
| 1b. Demotic Wire Script (Active LRU Macros) | `macro_lexicon.json` (256 Active + Archive) | `§X01..§X969` (e.g., `§X64`, `§X354`; 3 BPE tok/symbol) | Depth-1 macro expansion into core primitives; pinned core macros prevent LRU thrashing. |
| 2. Sacred Expanded Script (Canonical AST) | `codec.decode()` Output | `^§[A-Z][A-Z0-9_]*$` (e.g., `§GAIA_WARMTH_HARMONY`) | Zero lowercase or CJK corruption permitted; strictly enforced by unit test `test_17` across all files. |
| 3. Rosetta Audit Layer (Human Definitions) | `rosetta_layer.json` & `definitions` | Natural-language semantic specification per symbol | Used for human governance inspection, `audit_corpus_alignment.py`, and `eval_comprehension.py --cond C`. |
| Inline Structured Payload (`[k:v]`) | Quote-Aware Scanner (`_parse_payload_block`) | `[pair:"USD/JPY", stop:150.5, code:"007", note:"a, b"]` | Stateful character scanner preserving quoted strings, leading zeros, commas, brackets, and escapes. |

### 2.2 Conflict-Free LL(1) EBNF Grammar and Quote-Aware Stateful Payload Parser

To guarantee deterministic $\mathcal{O}(L)$ parsing without regex backtracking bugs or payload corruption, `v5.4.4` implements a conflict-free LL(1) grammar paired with a stateful character-level quote/escape scanner in `src/codec.py`:

```
packet = [ "@CTX:", identifier, "::" ], expr_node, { ( causal_arrow | "=>" | "!->" | ">" | "?" ), expr_node } ;
expr_node = ( "(", packet, ")" | "{", packet, "}" | "§", ( risc_id | macro_id | sacred_name ) ), [ "[", kv_pair, { ",", kv_pair }, "]" ] ;
kv_pair = identifier, ":", ( '"', { ( utf8_char - ( '"' | '\' ) ) | '\"' | '\\' }, '"' | float_lit | int_lit | bool_null_id ) ;
causal_arrow = "-[", [ "p:", float_val, "," ], ( "c:" | "cost:" ), cost_mantissa, "]->" ; (* cost_mantissa: exact 4 digits *)
```

The parser strictly distinguishes quoted strings (`"007"` → `str`) from unquoted numbers (`007` → `int`), isolates delimiters inside quotes (`note:"a, b"`, `expr:"x]y"`), and enforces explicit OOV policies (`on_oov="error" | "keep" | "nearest"` with an audit log in `EncodedWireResult.substitutions`, verified by tests `test_07`–`test_15`).

## 3. Formal Threat Model and Game-Based EUF-CMA Proof of Origin Attestation

### 3.1 Unobservable Internal-Counter Attestation Construction (Zero Timestamp Jitter)

To prevent an active Dolev-Yao adversary $\mathcal{A}$ from injecting forged commands or observing rejection-sampling loops under an adaptive Chosen-Message Attack (EUF-CMA), TOAI-elegant-lang embeds an origin tag directly into the 4-digit causal cost mantissa $c_n \in [0.0000, 0.9999]$. Session key $S_0 \in \{0,1\}^{256}$ is established via authenticated X25519 ECDH (with static host keys) and HKDF-SHA256, guaranteeing cross-session replay immunity; strict timestamp monotonicity $t_n > t_{n-1}$ enforces per-turn freshness and rejects clock skew at the receiver. Let $H_0 = \text{SHA256}(\texttt{"TOAI_TELEPATH_GENESIS_V541"} \,\|\, \text{SHA256}(\texttt{core_lexicon.json}) \,\|\, \text{SHA256}(\texttt{macro_lexicon.json}))$, and $H_{n-1} = \text{SHA256}(H_{n-2} \,\|\, \texttt{0x00} \,\|\, \text{wire}_{n-1})$. For monotonic timestamp $t_n > t_{n-1}$, sender and verifier compute the 16-bit value $Y_n \in \{0, \dots, 59{,}999\}$ for public confidence mantissa $P_n = \lfloor p_n \times 10^4 \rfloor \in \{0, \dots, 9999\}$ via an **internal, non-transmitted rejection counter** $r_n \in \{0, 1, \dots\}$:

$$r_n = \min\Big\{ r \in \mathbb{N}_0 : \big(P_n \oplus \text{trunc}_{16}(\text{HMAC-SHA256}(S_0,\, H_{n-1} \,\|\, \texttt{0x00} \,\|\, \text{dom}(t_n, r)))\big) < 60{,}000 \Big\} \tag{1}$$

where $\text{dom}(t_n, 0) = t_n$ and $\text{dom}(t_n, r) = t_n \,\|\, \text{uint8}(r)$ for $r \geq 1$. Verifiers evaluate:

$$T_n = \text{trunc}_{16}\big(\text{HMAC-SHA256}(S_0,\, H_{n-1} \,\|\, \texttt{0x00} \,\|\, \text{dom}(t_n, r_n))\big), \qquad \big((P_n \oplus T_n) \bmod 10{,}000\big) = C_n \tag{2}$$

Because bitwise XOR is a bijection on $\{0, \dots, 65{,}535\}$, each candidate $P_n \oplus T_n^{(r)}$ is uniform with single-draw acceptance probability $p_{\text{acc}} = 60{,}000/65{,}536 = 91.55\%$. In this geometric trials setting, the expected internal evaluations per turn is $\mathbb{E}[r_n + 1] = 1/p_{\text{acc}} = 65{,}536/60{,}000 \approx 1.09226 \leq 1.0923$, bounding total PRF oracle queries by $\mathbb{E}[Q_{\text{PRF}}] \leq 1.0923(Q + k)$. For $r_1 = 0$ (Table 3), $Y_1 = 9841 \oplus 9247 = 622 < 60{,}000$, yielding $C_1 = 622 \bmod 10{,}000 = 622 \implies c_1 = 0.0622$.

### 3.2 Formal Security Reduction (Game$_0$ → Game$_2$) and Congruence Cardinality Proof

**Theorem 1 (Exact EUF-CMA Forgery Bound Subsuming Direct Tag Guessing and Modular Aliases).** *Let $\mathcal{A}(\tau, Q, k)$ be an adaptive EUF-CMA adversary making at most $Q$ signing queries to MAC oracle $\mathcal{O}_{MAC}(S_0, \cdot)$ and outputting a $k$-turn unqueried forgery trajectory $\{(wire^*_{n+j}, t^*_{n+j}, P^*_{n+j}, C^*_{n+j})\}_{j=1}^{k}$ with $t^*_{n+j} > t^*_{n+j-1}$. Under the $(\tau', Q_{PRF}, \epsilon_{PRF})$-PRF security of HMAC-SHA256 (where $\mathbb{E}[Q_{PRF}] \leq 1.0923(Q + k)$) and $(\tau', \epsilon_{coll})$-collision resistance of SHA-256:*

$$\text{Adv}^{\text{EUF-CMA}}_{\text{TOAI}}(\mathcal{A}) \leq \left(\frac{6}{60{,}000}\right)^k + \epsilon_{\text{PRF}} + \epsilon_{\text{coll}} = 10^{-4k} + \epsilon_{\text{PRF}} + \epsilon_{\text{coll}} \qquad \left(P^{\text{unrej}}_{\text{forge}}(k) \leq \left(\frac{7}{65{,}536}\right)^k + \epsilon_{\text{PRF}} + \epsilon_{\text{coll}}\right) \tag{3}$$

*Proof.* We define three security games Game$_0$ → Game$_1$ → Game$_2$, letting $W_i$ denote $\mathcal{A}$'s winning event in Game$_i$:

- **Game$_0$ (Real EUF-CMA Experiment):** Challenger samples $S_0 \xleftarrow{\$} \{0,1\}^{256}$, answers $Q$ queries via Eqs. (1)–(2), and evaluates $\mathcal{A}$'s $k$-turn forgery. By definition, $\Pr[W_0] = \text{Adv}^{\text{EUF-CMA}}_{\text{TOAI}}(\mathcal{A})$.
- **Game$_1$ (Transcript Collision Abort):** Challenger aborts if two distinct wire histories ever produce matching chain digest $H = H'$. Any abort yields a SHA-256 collision within $\tau' \approx \tau + \mathcal{O}(Q + k)$, hence $|\Pr[W_0] - \Pr[W_1]| \leq \epsilon_{\text{coll}}$.

- **Game$_2$ (PRF Replacement):** We replace HMAC-SHA256$(S_0, \cdot)$ with a lazy uniform random function $\mathcal{R} : \{0,1\}^* \to \{0,1\}^{256}$. By PRF reduction over at most $Q_{\text{PRF}}$ evaluations, $|\Pr[W_1] - \Pr[W_2]| \leq \epsilon_{\text{PRF}}$. Because digests are collision-free (Game$_1$) and timestamps strictly monotonic, every input $X^{(r)}_{n+j} = H^*_{n+j-1} \,\|\, \texttt{0x00} \,\|\, \text{dom}(t^*_{n+j}, r)$ during forgery verification is strictly fresh (never previously queried).
- **Residue Class Analysis in Game$_2$:** For each forged turn $j \in \{1, \ldots, k\}$: (1) If $C^*_{n+j} \notin \{0, \ldots, 9999\}$, the LL(1) parser rejects deterministically ($\Pr = 0$). (2) If $C^*_{n+j} \in \{0, \ldots, 9999\}$, because $T \mapsto P^*_{n+j} \oplus T$ is a bijection and $\text{trunc}_{16}(\mathcal{R}(X^{(r)}_{n+j}))$ is uniform, the accepted draw $Y_{n+j} = P^*_{n+j} \oplus T^{(r_{n+j})}_{n+j}$ is uniform on $\{0, \ldots, 59{,}999\}$. The winning congruence set has exact cardinality $|\mathcal{W}(C^*_{n+j})| = 6$:

$$\mathcal{W}(C^*_{n+j}) = \left\{C^*_{n+j} + 10{,}000m : m \in \{0, \ldots, 5\}\right\} \subset \{0, \ldots, 59{,}999\} \implies \Pr\left[Y_{n+j} \in \mathcal{W}(C^*_{n+j})\right] = \frac{6}{60{,}000} = 10^{-4} \tag{4}$$

subsuming Direct Tag Guessing ($m = 0$, prob 1/60,000) and Modular Aliases ($m \in \{1, \ldots, 5\}$, prob 5/60,000). Across $k$ independent turns, $\Pr[W_2] = (6/60{,}000)^k = 10^{-4k}$. Without rejection sampling ($Y_{n+j} \in \{0, \ldots, 65{,}535\}$), residues $C^* \leq 5{,}535$ admit a 7th alias ($m = 6$), bounding forgery by $\max |\mathcal{W}_{\text{unrej}}|/65{,}536 = 7/65{,}536 \approx 1.068 \times 10^{-4}$. Non-transmitted $r_n$ adds $\approx 0.014$ $\mu$s CPU variance, leaking negligible timing side-channel signal ($\ll 10^4$ $\mu$s network/LLM jitter). □

**Table 3:** Reproducible Cryptographic Test Vector for Equation (2) (Verifiable via Python `hashlib`/`hmac`, $r_1 = 0$)

| Parameter / Verification Step | Canonical Hexadecimal / Decimal Value (Bitwise XOR Precedes mod 10,000) |
|---|---|
| Secret Seed $S_0$ (256-bit hex) | `7f83b1657ff1fc53b92dc18148a1d65dfc2d4b1fa3d677284addd200126d9069` |
| Session Anchor $H_0$ (SHA256) | `f3c515b25afe4f7ed4ca735b85fbfe50b083c7af4b8ec8c107c0e99438f02998` (`SHA256(b"TOAI_TELEPATH_GENESIS_V541")`) |
| Turn-1 Timestamp $t_1$ (UTF-8, $r_1 = 0$) | `2026-11-06T12:00:00Z` (HMAC-SHA256$(S_0, H_0 \,\|\, $ `b"\x002026-11-06T12:00:00Z"`$) \to T_1 =$ `0x241f` $= \mathbf{9247}$) |
| Step 1: 16-Bit Bitwise XOR | Given $p_1 = 0.9841 \implies P_1 = 9841$: $Y_1 = (P_1 \oplus T_1) = (9841 \oplus 9247) = \mathbf{622} < 60{,}000$ ($r_1 = 0$ accepted) |
| Step 2: Modulo 10,000 Reduction | $C_1 = Y_1 \bmod 10{,}000 = 622 \bmod 10{,}000 = \mathbf{622} \implies c_1 = C_1/10^4 = \mathbf{0.0622}$ |

## 4. Multi-AI Adversarial Audit, Non-Circular Corpus Alignment, and Holdout Disjointness

### 4.1 Full Disclosure of Multi-AI Adversarial Audit Findings (v5.4.1 → v5.4.4)

Table 4 discloses every defect identified during blind audits across OpenAI ChatGPT and Anthropic Claude execution sandboxes through release `v5.4.4`, all permanently verified by automated test suites.

**Table 4:** Summary of Multi-AI Adversarial Audit Findings and Resolutions in Canonical Release v5.4.4

| Audit Defect ID | Identified Vulnerability / Discrepancy | Deterministic Engineering Resolution in v5.4.4 & Unit Test |
|---|---|---|
| Defect 1: Wheel Packaging & Silent Dict Fallback | PyPI wheel omitted `lexicon/*.json`, and `ElegantCodec` silently initialized an empty dict `{}` when files were missing. | Packaged `src/lexicon/*.json` inside `pyproject.toml` wheel data; `ElegantCodec` raises `FileNotFoundError` / `RuntimeError` on missing/empty lexicons (`test_01`–`test_04`). |
| Defect 2: Payload Delimiter & `"007"` Coercion | Naive regex/comma splitting broke on `note:"a, b"`, `expr:"x]y"`, and coerced quoted `code:"007"` to integer 7. | Implemented stateful character-by-character quote/escape scanner in `_parse_payload_block()` and `_protect_quoted_spans()` (`test_07`–`test_10`, `test_15`). |
| Defect 3: Silent OOV Fallback to `§G01/§OMEGA` | Unregistered symbols were silently rewritten to `§G01` or `§OMEGA`, masking OOV failures in benchmarks. | Default `on_oov="error"` raises `UnknownSymbolError`; `"keep"` preserves symbol; `"nearest"` logs every fallback in `oov_substitutions` (`test_11`–`test_14`). |
| Defect 4: Naive Whitespace Token Estimator | Fallback token counter `r"\w+|\S"` undercounted BPE subwords when `tiktoken` was offline and hid estimation status. | Implemented CamelCase/snake_case/digit-chunk BPE subword estimator in `count_bpe_tokens()` returning explicit `is_estimated: bool` (`test_16`). |
| Defect 5: Lexicon & Corpus Symbol Typos (`GAiar`) | Mixed-case/corrupted identifiers (`§GAiar_WARMTH_HARMONY`, `§HARMONIC_ERECVERY_NEXUS`, CJK `U+822C/U+7684`, `_v2`) leaked into Q11/Q16 prompts. | Normalized all 256 primitives, 256 macros, Rosetta mappings, and 1,606 corpus records to strict `^§[A-Z][A-Z0-9_]*$` with zero undefined symbols (`test_17`). |
| Defect 6: Corpus Scenario Misalignment & Holdout Independence | 77.33% of legacy records in `corpus/dialogues.jsonl` had scenario labels mismatched with wire symbols; risk of circular evaluation if tested on repaired records. | Built non-circular auditor `src/audit_corpus_alignment.py` ($\kappa = 0.8611$ vs. human gold standard) AND constructed strictly disjoint $N = 112$ holdout set (`0.0%` overlap, `test_18`). |

### 4.2 Non-Circular Corpus Alignment ($\kappa = 0.8611$) and Strict Holdout Independence Proof

Historical logs (`corpus/dialogues.jsonl`, $N = 1{,}606$) contained severe misalignments: Recovery Pattern Leaks (FX scenarios containing only recovery ops) and Cross-Domain Bleed (parser debugging containing arbitrage macro `§X64`). We enforce two strict defenses in `src/audit_corpus_alignment.py`:

1. **Decoupled Independent Audit Ontology:** Completely isolated from generator dictionaries, the auditor evaluates 4 deterministic checks: (a) Plaintext leak rejection, (b) Cross-domain bleed blacklist, (c) Recovery leak check (> 50% recovery ops without domain anchors), and (d) Required domain anchor whitelist. Validated against a stratified human audit of $n = 60$ records (`corpus/human_audit_sample.csv`), the auditor achieved 22 True Matches, 34 True Mismatches, 2 FP, 2 FN ($p_o = 93.33\%$, chance $p_e = 52.00\%$, **Cohen's** $\kappa = 0.8611$, 95% CI: $[0.7296, 0.9926]$).
2. **Strict Disjointness Between Repaired Corpus and Evaluation Holdout Set (`--verify-holdout`):** To eliminate circularity between corpus repair (`--fix`) and evaluation, all comprehension benchmarks (Section 6) run on a **strictly disjoint Holdout Benchmark** (`corpus/holdout_comprehension.jsonl`, $N = 112 = 16$ items/domain $\times 7$ domains) built from 28 holdout-only scenarios. Running `python3 src/audit_corpus_alignment.py --verify-holdout` (and unit test `test_18`) proves: (i) Exact Wire Overlap: **0/112 (0.0%)**; (ii) Skeletal Wire Overlap (`@CTX` and `[k:v]` stripped): **0/112 (0.0%)**; (iii) Repair Scenario Overlap: **0/112 (0.0%)**; and (iv) Internal Unique Skeletal Wires: **112/112 (100.0%)**.

## 5. Empirical Evaluation I: Token Efficiency, Downstream Task Accuracy, and Header Amortization

In `benchmarks/fair_bench.py`, Demotic Wire is evaluated against alternative formats across representative multi-agent scenarios ($S_1$–$S_4$) and all $N = 112$ holdout tasks using standard Python `json.dumps(separators=(",", ":"))` conforming to JSON-RPC 2.0 / MCP schemas (`{"jsonrpc":"2.0", "method":..., "params":{...}}`):

**Table 5:** Independent Corpus Alignment Audit ($N = 1{,}606$ in `dialogues.jsonl`) and Disjoint Holdout Verification ($N = 112$ in `holdout_comprehension.jsonl`)

| Scenario Domain Category | Main $N$ | Pre-Fix OK | Post-Fix OK | Holdout $N$ (Overlap) | Primary Pre-Fix Failure Mode in Main Corpus |
|---|---|---|---|---|---|
| 1. `CODE_PARSER` (AST Bugfix) | 234 | 4 (1.71%) | **234 (100%)** | **16** (0.0% overlap) | Domain Bleed (`$X64 ARBITRAGE` macro) & Recovery Leak |
| 2. `FX` (Currency Hedge & Risk) | 245 | 5 (2.04%) | **245 (100%)** | **16** (0.0% overlap) | Recovery Pattern Leak (`REPLAY_GUARD` / `GRACEFUL_DRAIN`) |
| 3. `GAIA_HARMONY` (Co-Prosperity) | 228 | 98 (42.98%) | **228 (100%)** | **16** (0.0% overlap) | Recovery Pattern Leak & Plaintext Leak in legacy records |
| 4. `HYPOTHESIS` (Path Verify) | 192 | 0 (0.00%) | **192 (100%)** | **16** (0.0% overlap) | Missing Deductive/Bayesian Anchors & `$X64` Bleed |
| 5. `LATENT_SYNC` (Tensor Sync) | 241 | 29 (12.03%) | **241 (100%)** | **16** (0.0% overlap) | Missing Tensor/Vector/Manifold Anchors & Recovery Leak |
| 6. `RECOVERY` (Circuit Drain) | 245 | 128 (52.24%) | **245 (100%)** | **16** (0.0% overlap) | Domain Bleed (`ARBITRAGE` macro inside pure cluster drain) |
| 7. `TASK_BUDGET` (Quota Alloc.) | 221 | 18 (8.14%) | **221 (100%)** | **16** (0.0% overlap) | Recovery Pattern Leak (missing resource/quota primitives) |
| **Total / Aggregate Verified** | **1,606** | **364 (22.67%)*** | **1,606 (100%)** | **112 (0/112 = 0.0%)** | **Human Audit** ($n = 60$)**:** $p_o = 93.33\%$**, Cohen's** $\kappa = 0.8611$ |

*Note: 82 legacy records with free-form labels aligned prior to domain normalization ($282 + 82 = 364/1{,}606 = 22.67\%$); Holdout set is verified by `--verify-holdout` at 0/112 exact wire, skeletal wire, and repair-scenario overlap with `dialogues.jsonl`.

1. **Standard Lossless JSON Baselines:** (1) *Verbose A2A JSON* ($349.8 \pm 31.8$ tok; full NL prose + JSON AST; 346/391/348/314 tok); (2) *Structured Tool JSON* ($165.2 \pm 40.8$ tok; standard minified schema omitting prose; 145/225/159/132 tok); and (3) *Minified Key JSON* ($124.2 \pm 32.3$ tok; aggressive single-character keys `"c","s","a","p","t"`; 105/172/118/102 tok).
2. **LLMLingua-Style Perplexity Token Pruning ($\rho \in \{0.50, 0.35\}$) [4]:** Coarse-to-fine perplexity pruning on Verbose A2A. Punctuation deletion causes **Strict JSON Syntax Validity to drop to 0.0% (0/112)** at both ratios. Under heuristic AST recovery (using regex parameter boundary matching to salvage fields from pruned unparseable text), **Downstream Numeric & Task Execution Accuracy drops to 97.3% (109/112) at $\rho = 0.50$ (174.9 tok) and collapses to 61.6% (69/112) at $\rho = 0.35$ (122.4 tok)** due to deleted stop-loss thresholds and weights.
3. **Simplified Flat Template (`KEY=VAL`):** Requires 34.5 tok ($-90.1\%$), but retains only **25.0% (28/112) of numeric parameters** and achieves **0.0% (0/112) multi-branch causal execution accuracy** because intermediate steps, probabilities $(p, c)$, and fault edges (`!->`) are discarded.
4. **TOAI Demotic Wire (v5.4.4):** Encodes the full causal AST, all typed `[k:v]` parameters, and fallbacks in **73.2 BPE tokens/message** ($73.2 \pm 18.4$ tok; 59/100/71/63 tok), achieving **$-79.1\%$ vs. Verbose A2A**, **$-55.7\%$ vs. Structured JSON**, and **$-41.0\%$ vs. Minified JSON** with **100.0% (112/112) syntax validity, 100.0% numeric fidelity, and 100.0% task execution accuracy** (Table 6).

**Header Amortization (Break-Even Analysis):** Transmitting the 1,000-token dictionary header ($H = 1{,}000$) in zero-shot sessions breaks even when cumulative savings equal or exceed $H$: $N_{\text{break-even}} = \lceil H/(T_{\text{base}} - T_{\text{wire}}) \rceil$. For Verbose A2A ($\Delta_{\text{tok}} = 276.5$ tok/msg), $N^* = \lceil 1000/276.5 \rceil = \mathbf{4}$ messages, yielding $4 \times 276.5 = 1{,}106 \geq 1{,}000$ tokens saved (+106 net savings). Against Structured Tool JSON ($\Delta = 92.0$), $N^* = \lceil 1000/92.0 \rceil = \mathbf{11}$ msgs ($11 \times 92.0 = 1{,}012 \geq 1{,}000$). Under 90% prompt caching ($H_{\text{eff}} = 100$ tok), break-even occurs within **1 to 2 messages** (Table 7):

$$N^{\text{std}}_{\text{break-even}} = \left\lceil \frac{H}{T_{\text{baseline}} - T_{\text{wire}}} \right\rceil, \qquad N^{\text{cache (90\%)}}_{\text{break-even}} = \left\lceil \frac{0.10 \cdot H}{T_{\text{baseline}} - T_{\text{wire}}} \right\rceil \tag{5}$$

**Table 6:** Multi-Format BPE Token Efficiency vs. Downstream Task/AST Execution Accuracy (`python3 benchmarks/fair_bench.py`, $N = 112$ Tasks)

| Representation / Compression Method | o200k Tok | Token Reduction | Strict Syntax OK | Numeric Fidelity | Downstream Task Acc. | Operational & Structural Trade-Off Summary ($S_1/S_2/S_3/S_4$ Tok) |
|---|---|---|---|---|---|---|
| 1. Verbose A2A JSON (Baseline) | 349.8 | 0.0% (Base) | 100.0% | 100.0% | **100.0% (112/112)** | Full NL prose + JSON AST (346 / 391 / 348 / 314 tok). |
| 2. Structured Tool JSON | 165.2 | $-52.8\%$ | 100.0% | 100.0% | **100.0% (112/112)** | Standard lossless schema; repeats JSON keys (145 / 225 / 159 / 132 tok). |
| 3. Minified Key JSON (`"c","s"`) | 124.2 | $-64.5\%$ | 100.0% | 100.0% | **100.0% (112/112)** | Short keys; heavy brace/quote BPE cost (105 / 172 / 118 / 102 tok). |
| 4. LLMLingua Pruning ($\rho = 0.50$) [4] | 174.9 | $-50.0\%$ | **0.0%** (Broken) | 97.3% (109/112) | **97.3%** (Heuristic) | Breaks JSON parser; loses numeric params on 3/112 tasks. |
| 5. LLMLingua Pruning ($\rho = 0.35$) [4] | 122.4 | $-65.0\%$ | **0.0%** (Broken) | **61.6% (69/112)** | **61.6%** (Heuristic) | Still 1.67× larger than TOAI Wire, yet loses 38.4% of task parameters. |
| 6. Simplified Flat Template (Lossy) | 34.5 | $-90.1\%$ | 100.0% (Flat) | 25.0% (28/112) | **0.0% (0/112)** | Drops multi-step causal graph, weights $(p, c)$, and `!->` fallbacks. |
| **7. TOAI Demotic Wire (v5.4.4 Ours)** | **73.2** | **$-79.1\%$** | **100.0%** | **100.0% (112/112)** | **100.0% (112/112)** | **Best lossless text-API AST (59 / 100 / 71 / 63 tok).** |

**Table 7:** Session Dictionary Header Amortization / Break-Even Analysis (System Dictionary Header $H = 1{,}000$ BPE Tokens, Eq. 5)

| Comparison Target Baseline | Target Tok/Msg | Wire Tok/Msg | Saved / Msg | Break-Even (Std) | Break-Even (90% Prompt Cache) |
|---|---|---|---|---|---|
| Verbose A2A (Baseline) | 349.8 tok | 73.2 tok | **276.5 tok** | **4 msgs** ($4 \times 276.5 = 1{,}106 \geq 1{,}000$ tok) | **1 msg** (Effective header = 100 tok) |
| Structured Tool JSON (Lossless) | 165.2 tok | 73.2 tok | **92.0 tok** | **11 msgs** ($11 \times 92.0 = 1{,}012 \geq 1{,}000$ tok) | **2 msgs** (Effective header = 100 tok) |
| Minified Key JSON (Short Keys) | 124.2 tok | 73.2 tok | **51.0 tok** | **20 msgs** ($20 \times 51.0 = 1{,}020 \geq 1{,}000$ tok) | **2 msgs** (Effective header = 100 tok) |

## 6. Empirical Evaluation II: Large-Scale Disjoint Holdout Comprehension ($N = 112$)

### 6.1 Experimental Hygiene, Five Blind Conditions (A–E), and Statistical Methodology

We strictly separate deterministic round-trip codec losslessness (100.0%) from zero-shot receiver semantic comprehension (identifying the correct operational scenario among 4 choices at `temp` $= 0$ with `@CTX` and `[k:v]` stripped) across five paired conditions on the disjoint holdout benchmark (`corpus/holdout_comprehension.jsonl`, $N = 112$). Evaluators receive the standardized prompt: *"Given the following message and optional glossary, select the single most likely scenario from options A–D. Output ONLY the chosen letter: A, B, C, or D."* To ensure 100% deterministic reproducibility without external API rate limits or model version drift, our primary benchmark utilizes the self-contained `blind_semantic` evaluator (an offline, zero-temperature semantic classifier inspecting only the

rendered prompt text without metadata access; `benchmarks/eval_comprehension.py`), while the harness also supports commercial LLMs (Claude 3.5 Sonnet, GPT-4o, Gemini 2.0 Flash) under identical prompts:

- **Cond A (Compact Wire Skeleton Only):** Bare opcodes and causal arrows (`§X64 -[p:0.99, c:0.01]-> §G48 => §C20`; 274.6 tok).
- **Cond B (Sacred Expanded Names Only):** Expanded uppercase Sacred names (`§GAIA_WARMTH_EXCHANGE`; 315.3 tok, +40.7 tok vs. A).
- **Cond C (Sacred + Rosetta Glossary):** Cond B + natural-language definitions of wire symbols (595.3 tok).
- **Cond D (Shuffle Control – Negative Control): Strictly length-matched to Cond B (315.4 tok, diff 0.1 tok)**, randomly permuting Sacred symbol names across items via seeded derangement while preserving identical arrow/parenthesis syntax.
- **Cond E (Plaintext Upper Bound):** Concise natural-language summary (666.1 tok), establishing the task ceiling.

Confidence intervals are Wilson 95% CIs (Eq. 6), and paired condition differences on the same 112 items are evaluated via exact two-sided McNemar tests on discordant pairs ($b = n_{10}, c = n_{01}$), continuity-corrected $\chi^2_{\text{cc}} = (|b-c|-1)^2/(b+c)$, Odds Ratio $\text{OR} = c/b$, and Cohen's $h$:

$$\text{CI}_{\text{Wilson}} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \qquad p_{\text{McNemar}} = \min\left(1,\, 2\sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 0.5^{b+c}\right) \tag{6}$$

## 6.2 Large-Scale Disjoint Holdout Results ($N = 112$) and Paired McNemar Analysis

Table 8 and Table 9 report the empirical results on the $N = 112$ disjoint holdout (`logs/holdout_comprehension_report.json`):

- **Highly Significant Gain from Sacred Expansion (Cond A → Cond B, $p = 1.36 \times 10^{-9}$):** On bare Compact Wire (Cond A), accuracy is **44.6% (50/112)** (95% CI: [35.8%, 53.9%]; Routine Recovery 68.8%, Non-Recovery 40.6%). Expanding to Sacred names (Cond B) raises accuracy by +40.2 **percentage points to 84.8% (95/112)** (95% CI: [77.0%, 90.3%]; Routine Recovery **100.0%**, Non-Recovery **82.3%**). Discordant pairs ($b = 7, c = 52$) yield $\chi^2_{\text{cc}} = 32.81$, OR = 7.43, Cohen's $h = +0.878$, and exact $p = 1.36 \times 10^{-9}$ at an extra cost of only +40.7 tok/item (**+1.69%pt gain per 1,000 BPE tokens**).
- **Significant Gain from Rosetta Glossary (Cond B → Cond C, $p = 6.35 \times 10^{-3}$):** Injecting Rosetta definitions (Cond C) raises accuracy to **93.8% (105/112)** (95% CI: [87.7%, 96.9%]; Non-Recovery **92.7%**), resolving residual cross-domain ambiguities ($b = 1, c = 11$, OR = 11.00, $p = 6.35 \times 10^{-3}$) and statistically matching Plaintext ceiling E (**98.2%**, $p = 0.125$ ns).
- **Definitive Falsification of Structural Guessing (Cond B vs. Cond D, $p = 3.59 \times 10^{-17}$):** Under the length-matched Shuffle Control (Cond D, 315.4 tok), accuracy collapses to **24.1% (27/112)** (chance = 25.0%; $b = 72, c = 4, \chi^2_{\text{cc}} = 59.07, h = -1.315, p = 3.59 \times 10^{-17}$), proving comprehension is driven by the orthogonal semantics of Sacred vocabulary rather than syntax length or arrow topology.

**Table 8:** Receiver Blind Comprehension on Strictly Disjoint Holdout Benchmark ($N = 112$, 4-Choice Chance = 25.0%, `eval_comprehension.py --holdout`)

| Evaluation Condition | Prompt Tok | Holdout Acc. ($N = 112$) [Wilson 95% CI] | Routine Recovery ($n = 16$) [95% CI] | Non-Recovery ($n = 96$) [95% CI] | Gain / 1k Tok | Pilot Baseline ($n = 23$ Pre-Fix) & Analytical Role |
|---|---|---|---|---|---|---|
| **Cond A:** Compact Wire Only (`§S01..§X64`) | 274.6 | **44.6% (50/112)** [35.8%, 53.9%] | **68.8% (11/16)** [44.4%, 85.8%] | **40.6% (39/96)** [31.3%, 50.6%] | +0.64%pt | Pilot: 52.2% (12/23). Bare opcodes + causal arrows (`@CTX/[k:v]` stripped). |
| **Cond B:** Sacred Expanded (`§GAIA_WARMTH_*`) | 315.3 | **84.8% (95/112)** [77.0%, 90.3%] | **100.0% (16/16)** [80.6%, 100.0%] | **82.3% (79/96)** [73.5%, 88.6%] | **+1.69%pt** | Pilot: 69.6% (16/23). +40.2%pt over Cond A ($p = 1.36 \times 10^{-9}$). |
| **Cond C:** Sacred + Rosetta Glossary Definitions | 595.3 | **93.8% (105/112)** [87.7%, 96.9%] | **100.0% (16/16)** [80.6%, 100.0%] | **92.7% (89/96)** [85.7%, 96.4%] | +1.03%pt | +8.9%pt over Cond B ($p = 6.35 \times 10^{-3}$); statistically matches ceiling E ($p = 0.125$). |
| **Cond D:** Shuffle Control (Permuted Symbols) | 315.4 | **24.1% (27/112)** [17.1%, 32.8%] | **18.8% (3/16)** [6.6%, 43.0%] | **25.0% (24/96)** [17.4%, 34.5%] | −0.03%pt | Negative control: collapses to 25.0% chance ($p = 3.59 \times 10^{-17}$ vs. Cond B). |
| **Cond E:** Concise Plaintext Summary (Ceiling) | 666.1 | **98.2% (110/112)** [93.7%, 99.5%] | **100.0% (16/16)** [80.6%, 100.0%] | **97.9% (94/96)** [92.7%, 99.4%] | +0.98%pt | Positive control: natural-language scenario summary upper bound. |

**Table 9:** Paired Exact McNemar Significance Tests, Odds Ratios, and Cohen's $h$ Effect Sizes Across Conditions ($N = 112$ Disjoint Holdout)

| Paired Comparison | Discordant $(b, c)$ | $\chi^2_{\text{cc}}$ | Odds Ratio | Cohen's $h$ | Exact $p$-Value | Statistical & Architectural Interpretation |
|---|---|---|---|---|---|---|
| **Cond A vs. Cond B** | $(b = 7,\ c = 52)$ | 32.81 | 7.43 | +0.878 (Large) | $\mathbf{1.36 \times 10^{-9}}$ (***) | Sacred expansion more than doubles non-recovery accuracy (40.6% → 82.3%). |
| **Cond B vs. Cond C** | $(b = 1,\ c = 11)$ | 6.75 | 11.00 | +0.295 (Med) | $\mathbf{6.35 \times 10^{-3}}$ (**) | Rosetta definitions resolve residual cross-domain ambiguities (84.8% → 93.8%). |
| **Cond B vs. Cond D** | $(b = 72,\ c = 4)$ | 59.07 | 0.06 | −1.315 (Huge) | $\mathbf{3.59 \times 10^{-17}}$ (***) | Permuting symbol names destroys comprehension (84.8% → 24.1% chance). |
| **Cond C vs. Cond D** | $(b = 80,\ c = 2)$ | 72.30 | 0.03 | −1.610 (Huge) | $\mathbf{1.41 \times 10^{-21}}$ (***) | Full Tri-Script layer vs. structural-only negative control. |
| **Cond A vs. Cond D** | $(b = 38,\ c = 15)$ | 9.13 | 0.39 | −0.437 (Med) | $\mathbf{2.19 \times 10^{-3}}$ (**) | Compact prefixes (`§F`, `§G`) retain significant signal above permuted noise. |
| **Cond C vs. Cond E** | $(b = 1,\ c = 6)$ | 2.29 | 6.00 | +0.237 (Small) | 0.125 (ns) | Sacred + Rosetta (93.8%) approaches Plaintext ceiling (98.2%) within noise. |

## 7. Reproducibility Artifact & Self-Contained Verification Commands

All lexicons, the 1,606-record corpus, the 60-item human audit sample, the 112-item holdout benchmark, and all evaluation suites are available on PyPI (`pip install toai-elegant-lang==5.4.4`) and GitHub (https://github.com/PhenoX-AI-Alliance/TOAI-elegant-lang). Verified with Python 3.10+:

```
# 1. Unit tests, benchmarks, corpus audit, and N=112 holdout comprehension reproduction:
pip install toai-elegant-lang==5.4.4 && python3 -m unittest discover -s tests -v
python3 benchmarks/fair_bench.py
python3 src/audit_corpus_alignment.py --verify-holdout && python3 src/audit_corpus_alignment.py --score-sample corpus/human_audit_sample.csv
python3 benchmarks/eval_comprehension.py run --provider blind_semantic --cond ALL -d eval_prompts/holdout_112 --report-json logs/holdout_comprehension_report.json
# 2. Self-contained verification of Codec Round-Trip, Quote-Aware Payload, and Table 4 EUF-CMA Test Vector:
python3 -c '
import hashlib, hmac, sys; sys.path.insert(0, "src"); from codec import ElegantCodec
c = ElegantCodec(); stmt = "@CTX:FX_HEDGE :: §X64 [pair:\"USD/JPY\", code:\"007\", note:\"a, b\", expr:\"x]y\", stop:150.5] -[p:0.99, c:0.01]-> §PROTO_HEALTH_ATTEST
      => §GENTLE_RESOURCE_SHARING"
wire = str(c.encode(stmt, on_oov="error")); p = c.extract_payloads(wire)[0]
assert c.decode(wire, expand_macros=False) == stmt and p["code"] == "007" and p["note"] == "a, b" and p["expr"] == "x]y"
S0 = bytes.fromhex("7f83b1657ff1fc53b92dc18148a1d65dfc2d4b1fa3d677284addd200126d9069"); H0 = hashlib.sha256(b"TOAI_TELEPATH_GENESIS_V541").digest()
T1 = int.from_bytes(hmac.new(S0, H0 + b"\x002026-11-06T12:00:00Z", hashlib.sha256).digest()[:2], "big")
assert T1 == 9247 and ((9841 ^ T1) % 10000) == 622; print("ALL VERIFIED: Lossless AST, Quote-Aware, and EUF-CMA (T1=9247, C1=622)")
'
```

### References and Verifiable Artifacts


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